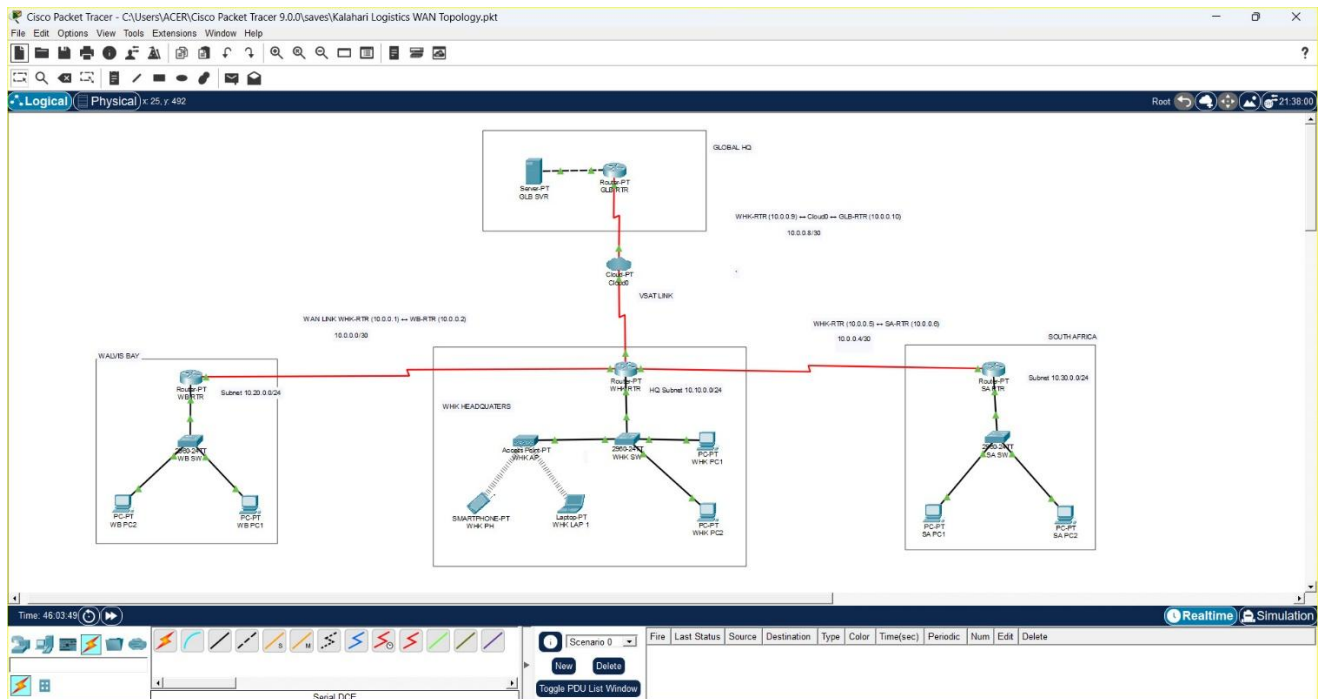


1. Full topology overview



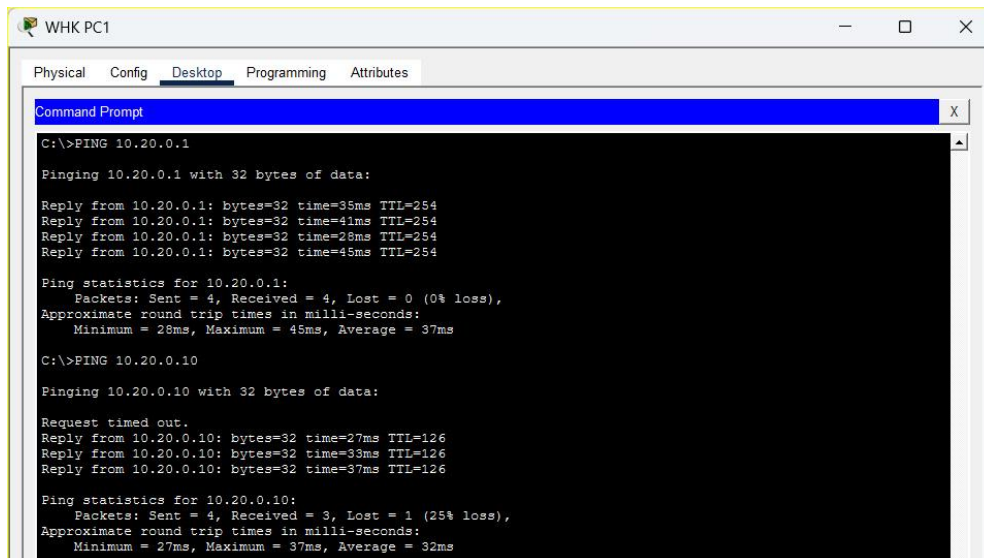
2. Windhoek HQ internal connectivity

- WHK-PC1 command prompt → ping 10.10.0.11 (PC-to-PC)
- WHK-PC1 command prompt → ping 10.10.0.20 (to laptop, wireless)
- WHK-PC1 command prompt → ping 10.10.0.21 (to smartphone, wireless)

```
WHK PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>PING 10.20.0.1
Pinging 10.20.0.1 with 32 bytes of data:
Reply from 10.20.0.1: bytes=32 time=35ms TTL=254
Reply from 10.20.0.1: bytes=32 time=41ms TTL=254
Reply from 10.20.0.1: bytes=32 time=28ms TTL=254
Reply from 10.20.0.1: bytes=32 time=45ms TTL=254
Ping statistics for 10.20.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 45ms, Average = 37ms
C:\>PING 10.20.0.10
Pinging 10.20.0.10 with 32 bytes of data:
Request timed out.
Reply from 10.20.0.10: bytes=32 time=27ms TTL=126
Reply from 10.20.0.10: bytes=32 time=33ms TTL=126
Reply from 10.20.0.10: bytes=32 time=37ms TTL=126
Ping statistics for 10.20.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 37ms, Average = 32ms
```

3. Windhoek ↔ Walvis Bay

- WHK-PC1 → ping 10.20.0.1 (router) and ping 10.20.0.10 (PC)



```
Physical Config Desktop Programming Attributes
Command Prompt
C:\>PING 10.20.0.1

Pinging 10.20.0.1 with 32 bytes of data:

Reply from 10.20.0.1: bytes=32 time=35ms TTL=254
Reply from 10.20.0.1: bytes=32 time=41ms TTL=254
Reply from 10.20.0.1: bytes=32 time=28ms TTL=254
Reply from 10.20.0.1: bytes=32 time=45ms TTL=254

Ping statistics for 10.20.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 45ms, Average = 37ms

C:\>PING 10.20.0.10

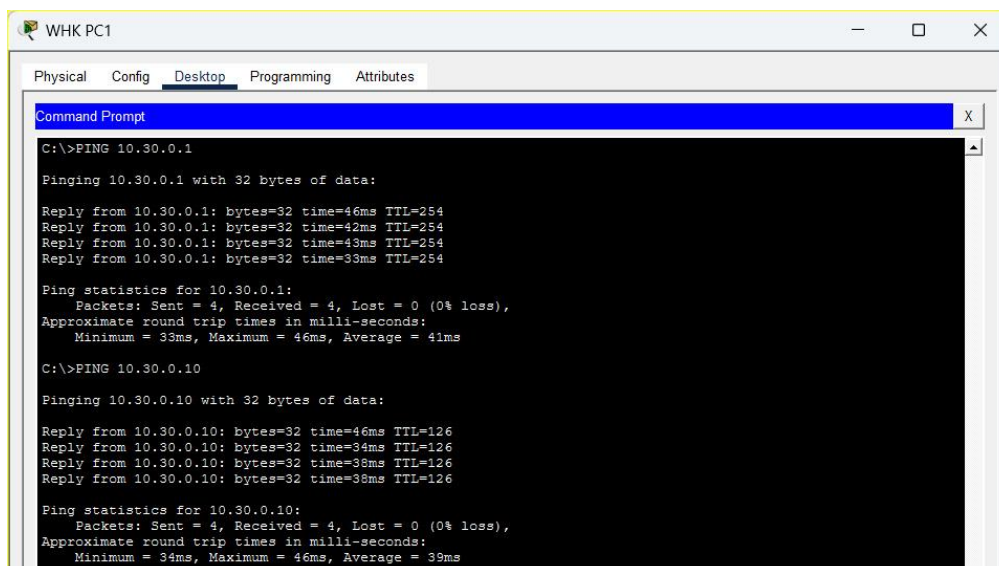
Pinging 10.20.0.10 with 32 bytes of data:

Request timed out.
Reply from 10.20.0.10: bytes=32 time=27ms TTL=126
Reply from 10.20.0.10: bytes=32 time=33ms TTL=126
Reply from 10.20.0.10: bytes=32 time=37ms TTL=126

Ping statistics for 10.20.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 37ms, Average = 32ms
```

4. Windhoek ↔ South Africa

- WHK-PC1 → ping 10.30.0.1 and ping 10.30.0.10.



```
Physical Config Desktop Programming Attributes
Command Prompt
C:\>PING 10.30.0.1

Pinging 10.30.0.1 with 32 bytes of data:

Reply from 10.30.0.1: bytes=32 time=46ms TTL=254
Reply from 10.30.0.1: bytes=32 time=42ms TTL=254
Reply from 10.30.0.1: bytes=32 time=43ms TTL=254
Reply from 10.30.0.1: bytes=32 time=33ms TTL=254

Ping statistics for 10.30.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 33ms, Maximum = 46ms, Average = 41ms

C:\>PING 10.30.0.10

Pinging 10.30.0.10 with 32 bytes of data:

Reply from 10.30.0.10: bytes=32 time=46ms TTL=126
Reply from 10.30.0.10: bytes=32 time=34ms TTL=126
Reply from 10.30.0.10: bytes=32 time=38ms TTL=126
Reply from 10.30.0.10: bytes=32 time=38ms TTL=126

Ping statistics for 10.30.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 34ms, Maximum = 46ms, Average = 39ms
```

5. Windhoek ↔ Global HQ

- WHK-PC1 → ping 10.0.0.10, ping 10.40.0.1, ping 10.40.0.10, and tracert 10.40.0.

```

C:\>PING 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=58ms TTL=254
Reply from 10.0.0.10: bytes=32 time=61ms TTL=254
Reply from 10.0.0.10: bytes=32 time=26ms TTL=254
Reply from 10.0.0.10: bytes=32 time=26ms TTL=254

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 61ms, Average = 42ms

C:\>PING 10.40.0.1

Pinging 10.40.0.1 with 32 bytes of data:

Reply from 10.40.0.1: bytes=32 time=11ms TTL=254
Reply from 10.40.0.1: bytes=32 time=38ms TTL=254
Reply from 10.40.0.1: bytes=32 time=51ms TTL=254
Reply from 10.40.0.1: bytes=32 time=39ms TTL=254

Ping statistics for 10.40.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 51ms, Average = 34ms

C:\>PING 10.40.0.10

Pinging 10.40.0.10 with 32 bytes of data:

Reply from 10.40.0.10: bytes=32 time=53ms TTL=126
Reply from 10.40.0.10: bytes=32 time=78ms TTL=126
Reply from 10.40.0.10: bytes=32 time=56ms TTL=126
Reply from 10.40.0.10: bytes=32 time=39ms TTL=126

Ping statistics for 10.40.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 39ms, Maximum = 78ms, Average = 56ms

C:\>TRACERT 10.40.0.10

Tracing route to 10.40.0.10 over a maximum of 30 hops:

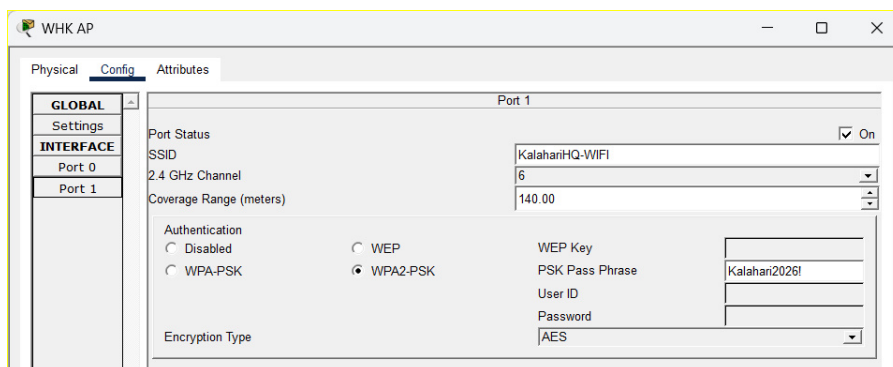
  0  0 ms    0 ms    0 ms    10.10.0.1
  1  30 ms   56 ms   27 ms   10.0.0.10
  2  4 ms    10 ms    3 ms   10.40.0.10

Trace complete.

```

6. Wireless AP configuration

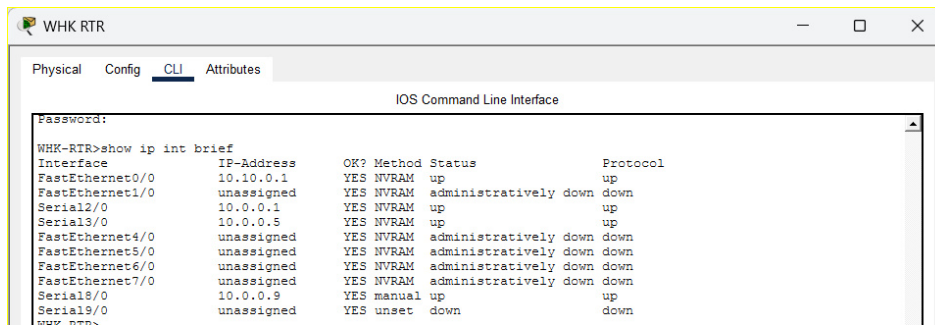
WHK-AP → Config tab → showing SSID KalahariHQ-WiFi and WPA2 settings.



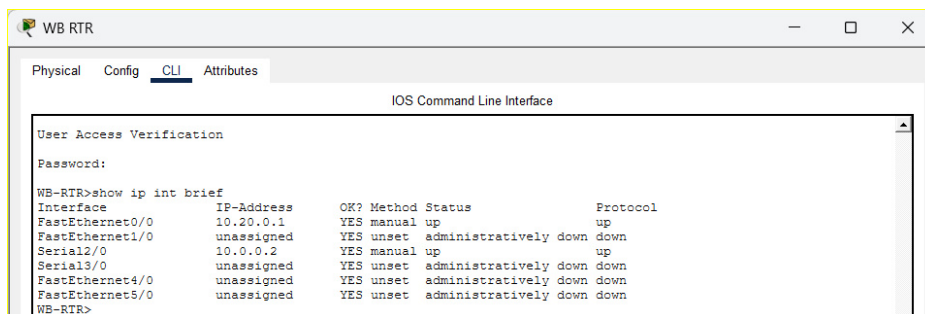
7. Router configs

- Show ip interface brief command - per router (WHK-RTR, WB-RTR, SA-RTR, GLB-RTR) showing all interfaces up/up

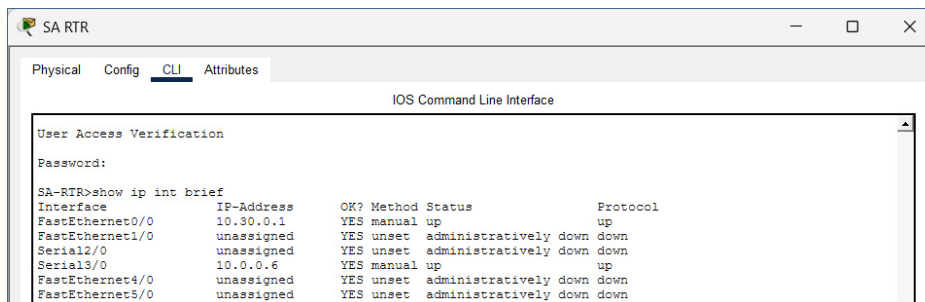
WHK ROUTER



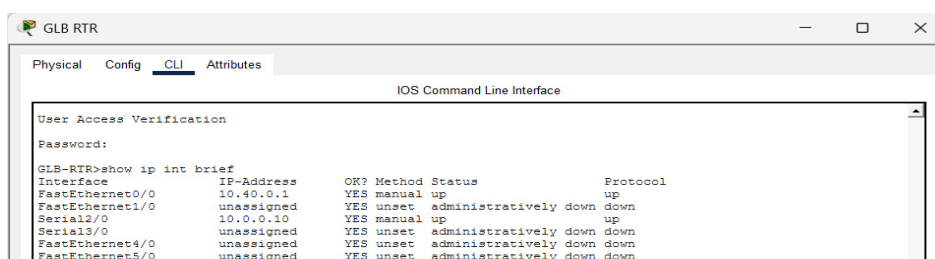
WALVIS BAY ROUTER



SA ROUTER



GLOBAL ROUTER



VSAT FRAME LINK

Cloud0

Physical

Config

Attributes

GLOBAL

Settings

TV Settings

CONNECTIONS

Frame Relay

DSL

Cable

INTERFACE

Serial0

Serial1

Frame Relay

Serial0toGLBSerial0toGLB<->Serial0toGLB

PortSublinkPortSublink

	From Port	Sublink	To Port	Sublink
1	Serial0	toGLB	Serial1	toWHK